

Solution

LIGHT -04

Class 10 - Science

- Optical centre of a thin lens is a point on the principal axis of lens such that a ray of light directed towards it, passes undeviated.
- The perpendicular distance between the path of emergent ray and the direction of incident ray is called lateral shift.
- Optical centre.
- A medium in which light travels comparatively slower than the other medium is called an optically denser medium.
- Refractive index has no units as it is a ratio of two similar physical quantities.
- Power of lens (in D) = $\frac{1}{\text{Focal length (in metre)}}$
 $P = \frac{1}{f}$
- Speed of light is minimum in medium A and maximum in medium C.
 - The focal length of a concave lens is negative, so its power is negative.
 - The focal length of a convex lens is positive, so its power is positive.
- Refractive index of a medium with respect to another medium is called relative refractive index.
- Magnification of a lens is to the ratio of the height of the image formed by lens to the actual height of object.
If h is the height of the object and h' the height of the image formed by lens, then magnification $m = \frac{h'}{h}$

If u and v are the distances of object and image

$$m = \frac{h'}{h} = \frac{v}{u}$$

- The refractive index of alcohol is more than water, therefore, alcohol is optically denser medium.
- A medium in which light travels comparatively faster than the other medium is called an optically rarer medium.
- Refractive index of a medium with respect to vacuum is called absolute refractive index.
- When a ray arrives from medium A, it moves towards normal in medium B. So, medium B has a higher optical density than medium A. This indicates that the refractive index of medium B relative to medium A's refractive index will be higher than 1.
 - Speed of light in rarer medium A (v_a) is greater than the speed of light in denser medium B (v_b) i.e., $v_a > v_b$
Thus, refractive index of B relative to A,
 $n = \frac{v_a}{v_b}$
 $\Rightarrow n > 1$
- Refraction can change path of light with change of medium.
- A concave lens is used to form an image formed is diminished and between focus and optical centre on the same side as that of the object.
- Refractive index of a medium can be defined as the ratio of Speed of light in vacuum to the Speed of light in the medium, i.e.,
Refractive index (of medium) = $\frac{\text{Speed of light in vacuum}}{\text{Speed of light in that medium}}$
- Medium P is optically denser than Q because in it the ray is bending more towards the normal (As $\angle r_p > \angle r_q$).
- A medium through which light can travel is called an optical medium.
- The working of a lens is based on the phenomenon of refraction of light.
- The ratio of the speed of light in the free space (c) to the speed of light in given medium (v) is called its refraction index.

$$n = \frac{c}{v}$$

It has no unit. Glass has more refractive index than water.

- We know that,

$$\text{Refractive index of ruby} = \frac{\text{Speed of light in air (or vacuum)}}{\text{Speed of light in ruby}}$$

$$\text{So, } 1.70 = \frac{3 \times 10^8}{\text{Speed of light in ruby}}$$

$$\text{or Speed of light in ruby} = \frac{3 \times 10^8}{1.7} \text{ ms}^{-1}$$

$$= 1.76 \times 10^8 \text{ ms}^{-1}$$

$$\text{Thus, speed of light in ruby is } 1.76 \times 10^8 \text{ ms}^{-1}$$

- The absolute refractive index of glass = 1.5

$$\text{Refractive index of diamond with respect to glass } n_{dg} = 1.6$$

Refractive index of glass $n_g = 1.5$

Refractive index of diamond $n_d = ?$

Refractive index of diamond with respect to glass

$$n_{dg} = \frac{n_d}{n_g}$$

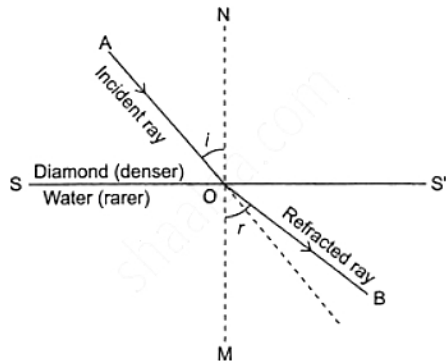
So, the refractive index of diamond $n_d = n_{dg} \times n_g$

or, $n_d = 1.6 \times 1.5 = 2.4$

Hence, the refractive index of diamond is 2.4.

24. The focal length of a lens increases and the power gets decreased.

25. a.



b. Refractive index of water w.r.t air is = 1.33

Similarly refractive index of diamond w.r.t. air is = 2.4

So,

Refractive index of water with respect to diamond is $= \frac{1.33}{2.4} = 0.55$

26. The angle of refraction will be 0° .

27. $n = \frac{c}{v}$ or $v = \frac{c}{n} = \frac{1}{1.5} \times 3 \times 10^8 \text{ ms}^{-1}$. It means, speed of light in glass is $\frac{1}{1.5}$ times the speed of light in air ($3 \times 10^8 \text{ ms}^{-1}$).

28. Refractive index, $n = \frac{\sin i}{\sin r}$, where i = angle of incidence in air and r = angle of refraction in glass slab.

29. Precautions are as follows:

- Fix the lens vertically in lens holder.
- Base of lens and white screen should be in a line with measuring scale.
- There should not be any obstacle in path of lens.
- Record the position of lens and screen when sharp image is formed.

30. A lens is a piece of transparent medium bounded by two spherical surfaces.

31. i. energy carried by the wave remains constant.

ii. wavelength decreases

32. The refractive index (n) of a medium is the ratio of the speed of light in vacuum (or air) to the speed of light in that medium.

$$n = \frac{\text{Speed of light in vacuum or air (c)}}{\text{Speed of light in that medium (u)}}$$

33. The convex mirror is used to form an image formed is diminished and between the pole and focus and behind the mirror.

34. One dioptre is defined as the power of lens having a focal length of 1 m.

35. The angle between the refracted ray and the normal at the point of incidence is called angle of refraction.

36. The bending of a ray of light falling obliquely on a surface when it passes from one medium to another is called refraction.

37. Lateral displacement depends on

- thickness of glass slab and
- angle of incidence.

38.

- Concave mirror is used when the image formed is virtual and erect.

- When an object is placed between the pole and focus than the image formed behind the mirror.

39. The power of a lens is defined as the reciprocal of the focal length. Lens power is measured in dioptres (D).

40. As per table, diamond has highest optical density (2.42). Medium with lowest optical density is air (1.0003)

41. i. Select a lens of small diameter.

ii. Select a distance object.

iii. Keep lights of the room on.

42. i. A convex lens

ii. A concave lens

43. $F=2\text{cm}$

$$P = \frac{1}{f}$$

$$P = \frac{1}{2} = 0.5 \text{ Diopetre}$$

44. Since lens is convex, therefore/is positive.

Given: $u = -20 \text{ cm}$, $f = +12 \text{ cm}$, $h = 3 \text{ cm}$, $v = ?$, $h' = ?$

Using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{We have, } \frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$$

$$\frac{1}{v} + \frac{1}{20} = \frac{1}{12}$$

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{5-3}{60} = \frac{1}{30}$$

$$v = 30 \text{ cm}$$

Since 'V' is positive, the image is located on the other side of the lens.

$$m = \frac{v}{u} = \frac{+30}{-20}$$

$$= -1.5$$

Since m is negative and greater than 1, the image is real, inverted and larger than the object.

$$m = \frac{h'}{h} = -1.5 = \frac{h'}{3}$$

$$\text{or } h' = -4.5 \text{ cm}$$

Thus the image is 30 cm from the convex lens, located on the other side of the lens. It is real, inverted and 4.5 cm high.

45. Given,

$$f = 30 \text{ cm}$$

$$u = -50 \text{ cm}$$

$$h = 6 \text{ cm}$$

Lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{30} - \frac{1}{50}$$

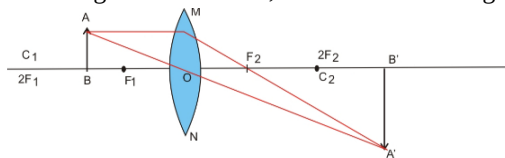
$$v = +75 \text{ cm}$$

$$m = \frac{v}{u} = \frac{h_i}{h}$$

$$\frac{75}{-50} = \frac{h_i}{6}$$

$$h_i = -9 \text{ cm}$$

The image formed is real, increased and enlarged.



46. i. In the above figure,

$$\angle N'OO' = \angle MO'O = \angle r = \text{angle of refraction}$$

lateral displacement = Distance O'B

ii. Given, refractive index from air to glass is $\frac{3}{2}$.

$$\text{i.e. } {}_a\mu_g = \frac{\mu_g}{\mu_a} = \frac{3}{2}$$

and we know that,

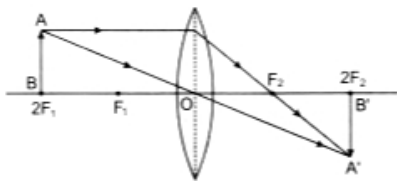
$$\Rightarrow {}_a\mu_g \times {}_g\mu_a = 1$$

$$\Rightarrow {}_g\mu_a = \frac{1}{{}_a\mu_g}$$

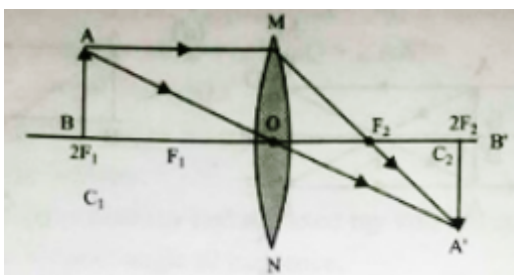
$$\Rightarrow {}_g\mu_a = \frac{1}{\left(\frac{3}{2}\right)} \Rightarrow {}_g\mu_a = \frac{2}{3}$$

i.e. refractive index of air for light going from glass to air is ${}_g\mu_a = \frac{2}{3}$

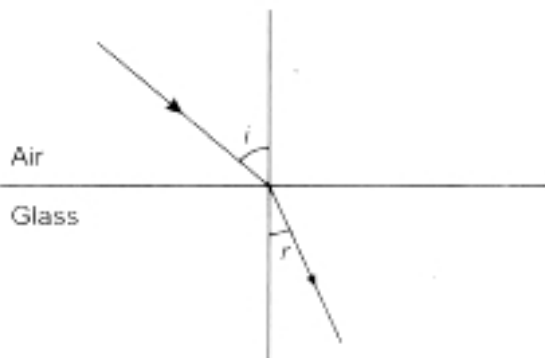
47. The object should be placed at $2F$.



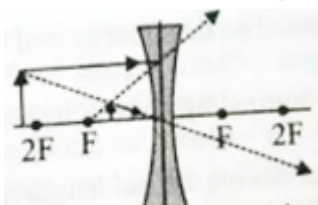
48. i. A ray of light parallel to the principal axis of the concave lens appears to be coming from focus after refraction through the lens.
 ii. A ray of light crossing through a principal focus, after refraction from a convex lens, will appear parallel to the principal axis.
 iii. A ray of light passing through the optical center suffers no refraction, that is, it passes undeviated. This is because, when it passes through the optical center, it is perpendicular to the curved surface of the lens. Hence, the angle of incidence is zero and therefore the angle of refraction is also zero.



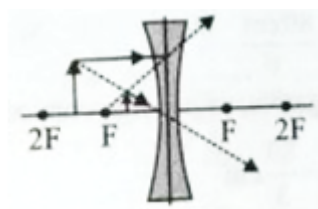
49.



50.



51.



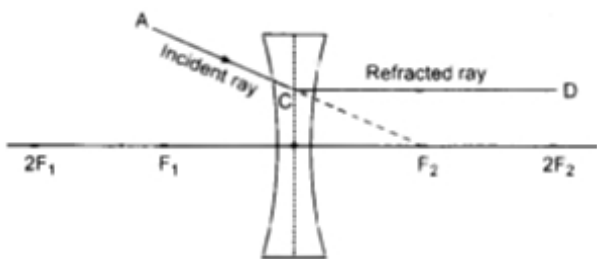
52.

$$53. f = \frac{1}{P} = \frac{1}{1.5} = 0.67 \text{ m}$$

$$f = 0.67 \text{ m}$$

Lens is converging.

54. In figure the refracted ray parallel to the principal axis. So, the incident ray must be appearing to meet at the principal focus of concave lens. To find the incident ray, F_2 is joined to C and produced as shown in the figure.



55. i. Concave lens.
 ii. The image formed at the centre of curvature of the mirror.
 iii. The concave mirror is used in the construction of shaving glass. Since concave mirrors magnify objects in focus and reflect real, almost 3-D images. Hence, used for shaving.

56. Luminous	Non-luminous
The objects which possess light of its own are called luminous objects.	The object which does not possess light of its own, but receives light from external source and scatters it to the surroundings is called non-luminous.
Examples: The Sun, the stars, lamp, candle, etc.	Examples: The Moon, a chair, a book, etc.

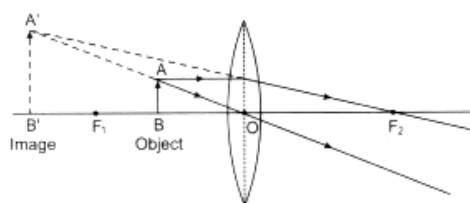
57. Refractive index of glass = $\frac{\text{Speed of light in air}}{\text{Speed of light in glass}}$

or, $1.50 = \frac{\text{Speed of light in air}}{\text{Speed of light in glass}}$

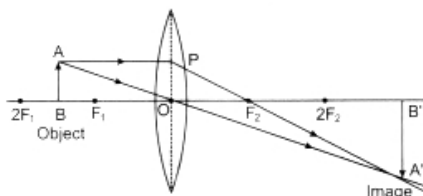
or, Speed of light in glass = $\frac{3 \times 10^8}{1.50} \text{ ms}^{-1}$

$= 2 \times 10^8 \text{ ms}^{-1}$

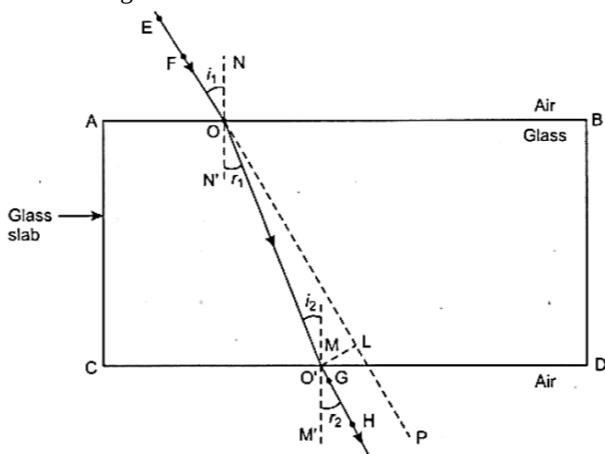
The speed of light in glass is $2 \times 10^8 \text{ ms}^{-1}$.



58.



59. The principal focus or the second focal point of a convex lens is a point on the principal axis of the lens such that the rays of light incident parallel to the principal axis, after refraction from the lens pass through it.
 60. It is shown in the ray diagram that light ray bends towards the normal when enters into the glass slab and away from the normal when emerges out in air.



61. Since the image is real and diminished, the lens must be convex and the object must be placed beyond 2F.

Given:

$v = +18 \text{ cm}$

$$\frac{h'}{h} = \frac{-4}{5} \dots [m = \frac{h'}{h} \text{ is negative for real image}]$$

f = ?

$$\text{We have, } m = \frac{h'}{h} = \frac{v}{u}$$

$$\frac{v}{u} = \frac{-4}{5}$$

$$\text{and } 4u = -5 \times 18 \text{ cm}$$

$$\text{Therefore, } u = -22.5 \text{ cm}$$

Also focal length is given by

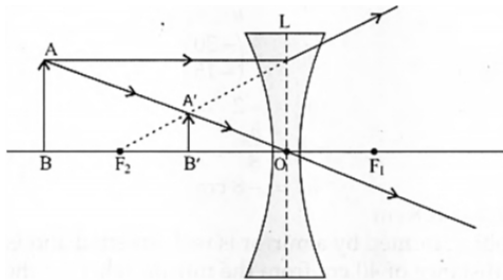
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{18} - \frac{1}{-22.5} = \frac{1}{18} + \frac{1}{22.5}$$

$$= \frac{9}{90}$$

$$\text{Therefore, } f = 10 \text{ cm}$$

62. It is a concave lens.



The power of a lens is given by the relation

$$\therefore P = \frac{1}{f(\text{in metre})}$$

$$P = 10 \text{ D}$$

$$\therefore f = \frac{1}{P} = \frac{1}{10} = 0.1 \text{ m}$$

or -10 cm, as lens is concave lens.

63. i. Refraction of light through a rectangular glass slab.

ii. The following are the laws of refraction of light:

- The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant, for the light of a given colour and for the given pair of media. This law is also known as Snell's law of refraction.

iii. Given, ${}_d\mu_g = 1.6$ and $\mu_g = 1.5$

$$\text{Refractive index of diamond with respect to glass} = \frac{\text{Absolute refractive index of diamond}}{\text{Absolute refractive index of glass} (\mu_g)}$$

$$\text{So, the absolute refractive index of diamond} = \text{Refractive index of diamond glass } ({}_d\mu_g) \times \text{Absolute refractive index of diamond glass } (\mu_g) \mu_d = 1.6 \times 1.5 = 2.4$$

64. Focal length of convex lens = 20 cm

Object distance, u = -15 cm

Using the lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} + \frac{1}{15}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{15}$$

$$v = -60 \text{ cm}$$

Four characteristics of the image formed by the lens are:

- Virtual
- Erect
- At a distance of 60 cm on the same side of the lens as the object.
- Enlarged image.

65. A concave lens always forms a virtual, erect image on the same side of the object.

Image-distance v = -10 cm; Focal length f = -15 cm; Object-distance u = ?

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or, } \frac{1}{u} = \frac{1}{v} - \frac{1}{f}$$

$$\frac{1}{u} = \frac{1}{-10} - \frac{1}{(-15)} = -\frac{1}{10} + \frac{1}{15}$$

$$\frac{1}{u} = \frac{-3+2}{30} = \frac{1}{-30}$$

or, $u = -30$ cm

Thus, the object-distance is 30 cm.

Magnification $m = \frac{v}{u}$

$$m = \frac{-10\text{cm}}{-30\text{cm}} = \frac{1}{3} \simeq +0.33$$

The positive sign shows that the image is erect and virtual. The image is one-third of the size of the object.

66. a. Concave lens.

b. Optical centre.

c. Given: $u = -30$ cm, $f = +20$ cm

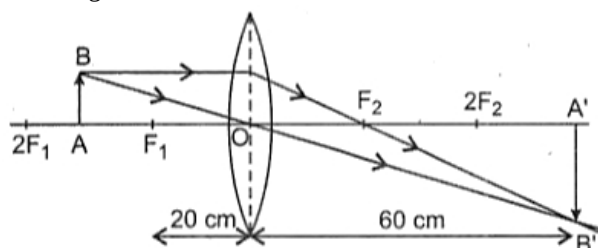
i. Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} + \frac{1}{-30} = \frac{3-2}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

The image is formed at a distance of 60 cm from the lens on the right side.



$$\text{ii. } \therefore m = \frac{v}{u} = \frac{60}{-30} = -2$$

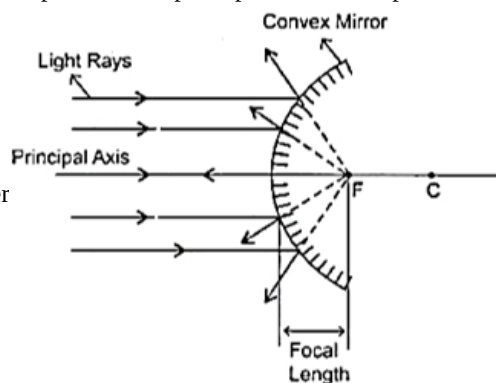
So, the image is inverted and double the size of the object.

iii. The image is real as $v > 0$, inverted and enlarged.

67. i. **Principal focus:** The principal focus of the diverging mirror is the point on the principal axis from which all incident rays appear to diverge from behind the mirror.

ii. **Focal length:** The distance between the pole and the principal focus of a spherical mirror is called the focal length.

Ray diagram in illustrate with answer



68. Power of lens is its ability to converge or diverge the light rays. A lens is small focal length converges or diverges the rays through a large angle and hence possess more power.

Power of lens in SI is measured by dioptre. Power of a lens is dioptre (D) is the reciprocal of focal length of lens in meters.

$$P = \frac{1}{f}$$

where P is in dioptre and f in meters. Powers are added algebraically. If lenses of power $P_1, P_2, P_3 \dots$ are placed in contact, their combined power is :

$$P = P_1 + P_2 + P_3 + \dots$$

69. **Laws of refraction:**

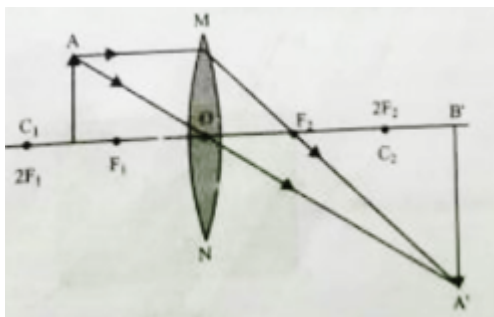
First law:- The incident ray, refracted ray and normal to the interface at the point of incidence lie in the same plane.

Second law:- The ratio of sine of angle of incidence with sine of angle of refraction is constant for a given pair of media.

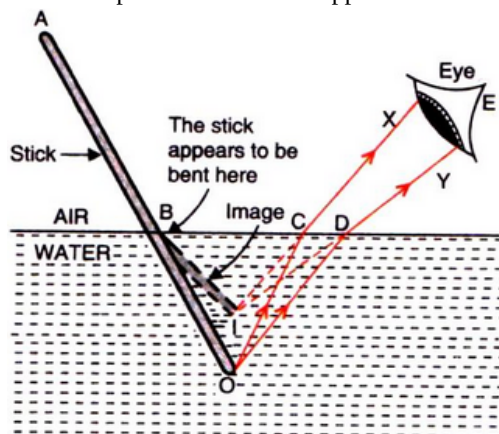
Absolute refractive index:- It is defined as the ratio of the velocity of light in a vacuum to the velocity of light in a medium. It is denoted by 'n'.

$$\text{Absolute refractive index } n = \frac{\text{velocity of light in vacuum}}{\text{velocity of light in medium}}$$

70.



71. A stick or a pencil half immersed in water at an angle appears bent due to refraction of light at the air-water surface. Figure shows a straight stick AO whose lower portion BO is immersed in water. It appears to be bent at point B in the direction BI. A ray of light OC coming from the lower end O passes from water into air at C and gets refracted away from the normal in the direction CX. Another ray OD gets refracted in the direction DY. The two refracted ray CX and DY, when produced backward, appear to meet at point I, nearer to the water surface than O. Similarly each part of the immersed portion of the stick raised. As a result immersed portion of the stick appears to be bent when viewed at an angle from outside.



72. Focal length = $\frac{1}{P} = \frac{1}{4} = 0.25 \text{ m}$ or 25 cm. It is a convex lens as power and focal length are positive.

Focal length, $f = +25 \text{ cm}$

Object distance, $u = -50 \text{ cm}$

Image distance, $v = ?$

Image height, $l = ?$

Using lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{25} = \frac{1}{v} - \frac{1}{(-50)}$$

$$v = +50 \text{ cm}$$

$$\text{Also, magnification, } m = -\frac{v}{u}$$

$$\text{Hence, } m = \frac{(50)}{-(-50)} = -1$$

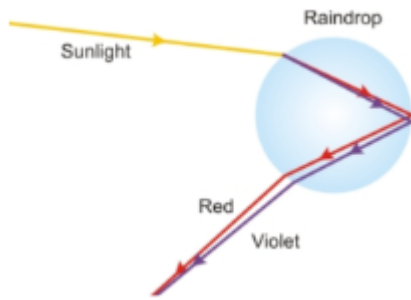
Image formed 50 cm in front of the lens. Due to a magnification of 1, the image is the same size as the object. The image formed is inverted as evidenced by the negative magnification.

73. If n number of lenses are placed in close contact, then the power of the combination of lenses is equal to algebraic sum of the powers of the individual lenses.

$$P = P_1 + P_2 + \dots + P_n$$

74. A rainbow is a band of colors. It is a natural spectrum appearing in the sky after rain. It is formed by the dispersion of sunlight by tiny water droplets in the sky. The water droplets act as tiny prisms. Light strikes the outer surface of the droplets and enters the droplet. Now the light ray strikes the inner surface of the droplet and reflects internally instead of refracting and moving out. Now the internally reflected ray strikes the opposite inner surface of the droplet and now refracts and moves out. This ray is dispersed

into rainbow's colors. Multiple refraction and internal reflection give rise to a rainbow.



75. i. Visible spectrum is the band of coloured components of a white light beam.
ii. Red light is scattered the least by air molecules and has longer wavelength. It travels the longest distance.
iii. The given setup will behave like a glass slab, resulting in recombination of the seven colours to produce white light.